

AP CALCULUS AB - UNIT 1

Guided Notes 1.4: Estimating Limit Values from Tables



Name: _____ Class: _____ Date: _____

Learning Objectives and Essential Question

Essential question: How can we select input values that approach the target from both sides?

- Select input values that approach the target from both sides.
- Estimate finite, infinite, and nonexistent limits from tabular patterns.
- Distinguish the center value from nearby evidence.
- Evaluate the reliability of a table, including rounding, spacing, and deceptive sampling.

Key Knowledge, Vocabulary, and Notation

Direction

Inputs less than the target estimate the left-hand limit; inputs greater than the target estimate the right-hand limit.

Closeness

Use several inputs increasingly close to the target, not one isolated row.

Stable digits

Report only digits that remain stable as inputs move closer.

Center value

A row at $x = c$ records $f(c)$, not the limit.

Evidence, not proof

A finite table supports an estimate but cannot by itself prove all nearby behavior.

Concept Development and Strategy

1. Build two columns of inputs: $c - 10^{-k}$ and $c + 10^{-k}$.
2. Compare both sides separately before forming a two-sided conclusion.
3. Use calculator precision high enough to avoid rounding a hole, jump, or oscillation into a false pattern.

Shared Representation / Data

x	3.9	3.99	3.999	4.001	4.01	4.1
$f(x)$	7.81	7.9801	7.998001	8.002001	8.0201	8.21

x	-2.01	-2.001	-1.999	-1.99
$g(x)$	4.98	4.998	-0.997	-0.970

Worked Example: Finite estimate

Estimate the limit from the table.

x	3.9	3.99	3.999	4.001	4.01	4.1
$f(x)$	7.81	7.9801	7.998001	8.002001	8.0201	8.21

Answer and Reasoning

Both sides approach 8. The stable digits support $\lim_{x \rightarrow 4} f(x) = 8$.

Worked Example: One-sided disagreement

Analyze the table.

x	-2.01	-2.001	-1.999	-1.99
$g(x)$	4.98	4.998	-0.997	-0.970

Answer and Reasoning

From the left, values approach 5; from the right, they approach -1. Therefore the two-sided limit does not exist.

Worked Example: Hidden center value

A table includes $f(2) = 100$, while rows at 1.99, 1.999, 2.001, 2.01 are near 3. What should be concluded?

Answer and Reasoning

The nearby evidence supports $\lim_{x \rightarrow 2} f(x) = 3$, while $f(2) = 100$. The center row does not determine the limit.

Guided Practice and Discussion

Directions

Show the mathematical setup, preserve exact values unless a decimal is requested, and justify existence or interpretation statements with the appropriate definition or theorem.

1. Use the first shared table to estimate $\lim_{x \rightarrow 4^-} f(x)$.

2. Use the first shared table to estimate $\lim_{x \rightarrow 4^+} f(x)$.

3. Create four suitable x -values to estimate a two-sided limit at $x = 7$.

4. A table uses only $x = 1.1, 1.01, 1.001$. Which limit direction can it directly estimate at $x = 1$?

5. A calculator rounds all outputs to two decimals and reports 1.00 on both sides. Explain why this does not prove the limit is exactly 1.

6. A table samples $f(x) = \sin(\pi/(x - 1))$ only at $x = 1 + 1/n$, where every output is 0. Why is the table deceptive?

7. Give two functions that agree at every input in the finite table $x = \pm 0.1, \pm 0.01, \pm 0.001$ but have different limits at 0.

AP Reasoning Check

Multiple-Choice Check

Choose the best answer. Explain why at least one distractor is incorrect during class discussion.

1. A table near $x = 5$ has outputs approaching -2 from both sides, while the center row is 17. Which conclusion is best?

- (A) $\lim_{x \rightarrow 5} f(x) = 17$ (B) $\lim_{x \rightarrow 5} f(x) = -2$
(C) The limit DNE. (D) No conclusion is possible.

2. Which input list is best for estimating $\lim_{x \rightarrow 3} f(x)$?

- (A) 0, 1, 2, 3 (B) 2.9, 2.99, 3.01, 3.1
(C) 3, 4, 5, 6 (D) $-3, -2, -1, 0$

3. A calculator table for $h(x) = \frac{e^x - 1}{x}$ gives 0.99950, 0.99995, 1.00005, 1.00050 near $x = 0$. The best estimate is

- (A) 0 (B) 0.5
(C) 1 (D) DNE

Common Errors and AP Precision

- Keep the value of a function at a point separate from its limiting behavior near that point.
- State one-sided behavior explicitly whenever a two-sided limit or continuity claim depends on agreement from both sides.
- A complete AP justification names the relevant definition, condition, or theorem rather than reporting only a numerical result.

Exit Ticket

Explain how two interlaced sequences can disprove a proposed table-based limit.

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